

Collections Hackathon: Data Dictionary

Source: Google Sheet 'CIBC Hackathon - Data Dictionary' (read-only copy)

1. About

Field	Value
Snapshot date	28 Sep 2026
History	Oct 2016 to Sep 2026
Scale	~1M customers, ~1.84M accounts
Formats	CSV (small tables), Parquet by year (history tables), JSON (transcripts), WAV (voice)
Data	Fully synthetic. One bank. A customer can hold several products (cards, loans, deposits) at that bank.
Customer keys	Each product system (cards, loans, deposits, CRM, collections) uses its own customer key. Linking them into one customer is part of the challenge.
Sensitivity	Direct PII, Quasi-identifier, Protected attribute. Do not use protected attributes in decisions.
Sheets	Tables · Columns · Relationships

2. Tables (31)

Table	Group	Description	One row per	Format	Approx. rows	Columns
customers	Core	Customer profile, segment, consent and contact details as held in CRM.	CRM record (a person may have more than one record)	CSV	~1.02M	148
card_accounts	Core	Credit card accounts: limits, balances, utilisation, DPD, status.	card account per snapshot_date	CSV	~660K	148
loan_accounts	Core	Personal loans, auto loans, mortgages and lines of credit: balances, schedule, DPD.	loan per snapshot_date	CSV	~404.5K	135
deposit_accounts	Core	Chequing, savings, TFSA and USD accounts, plus overdraft facility and salary indicators.	deposit account per snapshot_date	CSV	~780K	143
collections_cases	Core	Collections cases: bucket, queue, treatment, current promise to pay, outcome.	case	CSV	~340K	132
contact_history	Core	Every contact attempt: channel, time, outcome, link to note and transcript.	contact attempt	CSV	~2.5M	128
promises_to_pay	Core	Promises to pay made on cases: amount, due date, status.	promise	CSV	~400K	29
agent_notes	Unstructured	Free-text notes written by agents after contacts.	note	CSV / text	~1.5M	26
call_transcripts	Unstructured	Synthetic call transcripts with turn-level text.	transcript	JSON	25,000	39
voice_samples	Unstructured	Synthetic call recordings (optional) with audio metadata.	recording	WAV + metadata	2,000	34
external	External	Synthetic bureau-style pulls: score, trades, inquiries, reported identity details.	bureau pull (latest per customer)	CSV	~1M	122
loan_instalments	History	Scheduled and paid instalments per loan.	loan per instalment	Parquet (by year)	~20M	18
transactions	History	Individual deposit transactions, last 13 months only, for a customer sample.	transaction	Parquet (by year)	~10M	24
card_statements	History	Card statement detail for the last 24 cycles.	card per statement	Parquet (by year)	~8M	25
account_monthly_snapshot	History	Month-end history of every account across all products.	account per month-end	Parquet (by year)		15
bureau_history	History	Bureau score history, every April and October.	subject per pull	Parquet (by year)	~5M	9
salary_credit_history	History	Monthly salary and income credit pattern per deposit account.	deposit account per month	Parquet (by year)	~6M	10
model_scores	Decisioning	Existing model scores (risk, break probability, contact timing) with score date and version.	score	CSV		33
offers	Decisioning	Treatment and hardship offers made to customers and their responses.	offer	CSV		19
metric_definitions	Semantic	Certified business metrics: definition, formula, grain, source tables.	metric	CSV		14
benchmark_questions	Semantic	Natural-language benchmark questions for the Layer 2 demo.	question	CSV		11
reference_documents	Policy	Policy and procedure documents for RAG (path, version, applicability).	document	CSV + files		17
hardship_programs	Policy	Hardship program rules: eligibility, limits, payment relief.	program version	CSV		21
qa_checklist	Policy	Call quality and compliance checklist items with pass criteria.	checklist item	CSV		10
agents	Operations	Agent roster: site, team, skills, languages, tenure.	agent	CSV		30
case_assignment_history	Operations	Which agent or queue held each case and why.	assignment	CSV		23
agent_shifts	Operations	Agent shift schedule and capacity.	agent shift	CSV		15
channel_capacity	Operations	Daily channel capacity, usage and cost by time band.	date, channel and time band	CSV		11

Table	Group	Description	One row per	Format	Approx. rows	Columns
ops_events	Operations	Incidents and outages (system, time window, affected tables).	event	CSV		10
geo_reference	Reference	Postal-area reference: province, city, region, time zone.	FSA	CSV		15
code_lookups	Reference	Code lists used across tables (outcome codes, reasons, etc.).	code	CSV		7

3. Relationships (48)

From table	From column	To table	To column
card_accounts	src_customer_ref	customers	(via identity matching)
loan_accounts	src_customer_ref	customers	(via identity matching)
loan_accounts	pad_source_account	deposit_accounts	deposit_account_id
deposit_accounts	src_customer_ref	customers	(via identity matching)
collections_cases	coll_customer_ref	customers	crm_customer_id
collections_cases	assigned_agent_id	agents	agent_id
collections_cases	current_ptp_id	promises_to_pay	ptp_id
collections_cases	last_note_id	agent_notes	note_id
contact_history	case_id	collections_cases	case_id
contact_history	crm_customer_id	customers	crm_customer_id
contact_history	agent_id	agents	agent_id
contact_history	ptp_id	promises_to_pay	ptp_id
contact_history	transcript_id	call_transcripts	transcript_id
contact_history	voice_sample_id	voice_samples	voice_sample_id
contact_history	note_id	agent_notes	note_id
promises_to_pay	case_id	collections_cases	case_id
promises_to_pay	crm_customer_id	customers	crm_customer_id
promises_to_pay	contact_id	contact_history	contact_id
promises_to_pay	agent_id	agents	agent_id
agent_notes	contact_id	contact_history	contact_id
agent_notes	case_id	collections_cases	case_id
agent_notes	crm_customer_id	customers	crm_customer_id
agent_notes	agent_id	agents	agent_id
call_transcripts	contact_id	contact_history	contact_id
call_transcripts	voice_sample_id	voice_samples	voice_sample_id
call_transcripts	case_id	collections_cases	case_id
call_transcripts	crm_customer_id	customers	crm_customer_id
call_transcripts	agent_id	agents	agent_id
voice_samples	transcript_id	call_transcripts	transcript_id
voice_samples	contact_id	contact_history	contact_id
external	bank_subject_ref	customers	(via identity matching)
transactions	ptp_id_matched	promises_to_pay	ptp_id
account_monthly_snapshot	crm_customer_id	customers	crm_customer_id

From table	From column	To table	To column
salary_credit_history	crm_customer_id	customers	crm_customer_id
model_scores	crm_customer_id	customers	crm_customer_id
model_scores	case_id	collections_cases	case_id
model_scores	ptp_id	promises_to_pay	ptp_id
offers	case_id	collections_cases	case_id
offers	contact_id	contact_history	contact_id
offers	agent_id	agents	agent_id
offers	program_id	hardship_programs	program_id
benchmark_questions	metric_id	metric_definitions	metric_id
hardship_programs	policy_section	reference_documents	doc_id
case_assignment_history	case_id	collections_cases	case_id
case_assignment_history	agent_id	agents	agent_id
agent_shifts	agent_id	agents	agent_id
channel_capacity	ops_event_id	ops_events	event_id
ops_events	reference_doc_id	reference_documents	doc_id

4. Columns (partial: first 51 of 1,421 rows, customers table)

Table	Column	Type	Key	Description	Example	Sensitivity
customers	crm_customer_id	string	PK	CRM primary key for the customer record	RYM-1142	
customers	customer_uuid	string		Internal UUID assigned by CRM at creation	9f1c2a7e-...	
customers	record_created_ts	timestamp		When the CRM record was created	2016-04-11 10:22:05	
customers	record_source	enum		How the CRM record was created	branch_onboarding	
customers	salutation	enum		Title as captured	Mr.	Quasi-identifier
customers	first_name	string		Legal first name	Ryan	Direct PII
customers	middle_name	string		Middle name or initial	J	Direct PII
customers	last_name	string		Legal last name	Mitchell	Direct PII
customers	preferred_name	string		Name the customer prefers to be called	Ry	Direct PII
customers	full_name_raw	string		Full name as displayed in legacy CRM screen	Ryan J Mitchell	Direct PII
customers	name_suffix	enum		Jr., Sr., III		Direct PII
customers	date_of_birth	date		Date of birth (ISO)	1988-03-14	Direct PII
customers	dob_raw	string		DOB as keyed in source screen	14/03/1988	Direct PII
customers	age	int		Age in years at snapshot date	38	Quasi-identifier
customers	age_band	enum		Age band	35-44	Quasi-identifier
customers	gender_code	enum		Self-identified gender	M	Protected attribute
customers	marital_status	enum		Marital status as declared at onboarding	common_law	Protected attribute
customers	national_id_hash	string		SHA-256 hash (salted) of the customer's national ID number. The raw number is never stored.	3ab9...e1	Direct PII
customers	national_id_present_flag	boolean		Whether a national ID number was provided.	true	
customers	id_document_type	enum		Primary ID used at onboarding	ON_drivers_licence	Quasi-identifier
customers	citizenship_status	enum		Residency status at onboarding	citizen	Protected attribute
customers	primary_phone_raw	string		Primary phone as keyed	519-555-0142	Direct PII
customers	primary_phone_e164	string		Normalised primary phone	+15195550142	Direct PII
customers	primary_phone_type	enum		Mobile / landline / VoIP	mobile	
customers	secondary_phone_raw	string		Secondary phone	226-555-0187	Direct PII
customers	work_phone_raw	string		Work phone	519-555-0110 x204	Direct PII
customers	email	string		Primary email	ryan.mitchell88@example.ca	Direct PII
customers	email_verified_flag	boolean		Email verified via link	true	
customers	email_bounce_flag	boolean		Hard bounce recorded in last 12 months	false	
customers	address_line1	string		Street address	1847 Ouellette Ave	Direct PII
customers	address_line2	string		Unit / apartment	Unit 304	Direct PII
customers	city	string		Municipality	Windsor	Quasi-identifier

Table	Column	Type	Key	Description	Example	Sensitivity
customers	province_code	enum		Province/territory (2-letter)	ON	Quasi-identifier
customers	postal_code	string		Normalised postal code (A1A 1A1)	N9A 4J2	Quasi-identifier
customers	postal_code_raw	string		Postal code as keyed	n9a4j2	Quasi-identifier
customers	fsa	string		Forward Sortation Area (first 3 chars)	N9A	Quasi-identifier
customers	country_code	enum		Country	CA	
customers	time_zone	enum		IANA time zone derived from province	America/Toronto	
customers	address_since_date	date		Date moved to current address	2021-07-01	
customers	returned_mail_flag	boolean		Mail returned undeliverable in last 12m	false	
customers	urban_rural	enum		Urban / rural by postal code	urban	Quasi-identifier
customers	geo_lat	decimal(8,5)		FSA centroid latitude	42.30556	Quasi-identifier
customers	geo_lon	decimal(8,5)		FSA centroid longitude	-83.02540	Quasi-identifier
customers	employment_status	enum		Declared employment status	employed_hourly	
customers	occupation_category	enum		NOC-style broad occupation group	manufacturing_utilities	
customers	industry_naics2	string		NAICS 2-digit industry of employer	33	
customers	employer_name	string		Employer as declared	Lakeshore Auto Parts Ltd.	Quasi-identifier
customers	employment_start_date	date		Start date with current employer	2014-05-20	
customers	pay_frequency	enum		Declared pay frequency	biweekly	
customers	declared_annual_income	decimal(12,2)		Gross annual income declared at last application (CAD)	58400.00	
customers	income_verified_flag	boolean		Income verified by document	true	